

Mathematical Olympiad Challenges

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Problems taken from the book by the same name as the title, by Andreescu and Gelca.

Chapter 1.9 — Problem 1

Prove that

$$\frac{\sin x}{\cos x} + \frac{\sin 2x}{\cos^2 x} + \cdots + \frac{\sin nx}{\cos^n x} = \cot x - \frac{\cos(n+1)x}{\sin x \cos^n x}$$

Solution

The heart of the problem is to consider the related sequences,

$$a_k = \frac{\sin kx}{\cos^k x}, \quad b_k = \frac{\cos kx}{\cos^k x}.$$

We apply the cosine addition formula on b_k ,

$$\begin{aligned} b_k &= \frac{\cos[(k-1)x] \cos x - \sin[(k-1)x] \sin x}{\cos^k x} \\ &= b_{k-1} - a_{k-1} \tan x \end{aligned}$$

which implies that $a_k \tan x = b_k - b_{k+1}$. Indeed the LHS of our identity now telescopes down to,

$$\begin{aligned} a_1 + a_2 + \cdots + a_n &= (b_1 - b_{n+1}) \cot x \\ &= \cot x - \frac{\cos(n+1)x}{\sin x \cos^n x}. \blacksquare \end{aligned}$$

Chapter 1.9 — Problem 3

Let n be a positive integer and a a real number, such that a/π is an irrational number. Compute

$$\frac{1}{\cos a - \cos 3a} + \frac{1}{\cos a - \cos 5a} + \cdots + \frac{1}{\cos a - \cos(2n+1)a}$$

Solution

Motivated by the sum to product formulae we notice that

$$\begin{aligned}\cos a - \cos(2k+1)a &= \cos[(k+1)a - ka] - \cos[(k+1)a + ka] \\ &= 2\sin[(k+1)a]\sin(ka).\end{aligned}$$

So we wish to compute

$$S = \sum_{k=1}^n \frac{1}{2\sin[(k+1)a]\sin(ka)}.$$

If one looks for a function that might telescope into the general term of this series then we may figure out that

$$\begin{aligned}\frac{\cos(k+1)a}{\sin(k+1)a} - \frac{\cos(ka)}{\sin(ka)} &= \frac{\cos[(k+1)a]\sin(ka) - \cos(ka)\sin[(k+1)a]}{\sin(ka)\sin[(k+1)a]} \\ &= \frac{\sin(-a)}{\sin(ka)\sin(k+1)a}.\end{aligned}$$

Armed with this fact it is clear that we can re-write our sum as

$$S = -\frac{1}{2\sin a} \sum_{k=1}^n [\cot[(k+1)a] - \cot(ka)]$$

which telescopes down to

$$S = \frac{1}{2\sin a} (\cot a - \cot(n+1)a). \blacksquare$$

Chapter 3.2 — Problem 2

Show that each of the numbers

$$\frac{107811}{3}, \frac{110778111}{3}, \frac{111077781111}{3}, \dots$$

is the cube of a positive integer.

Solution

First we try to characterise the pattern given in these numbers. Let the k -th number in this list be n_k , then

$$3n_k = \underbrace{11\dots1}_{k \text{ times}} 0 \underbrace{77\dots7}_{k \text{ times}} 81 \underbrace{11\dots1}_{k \text{ times}}.$$

After looking at small cases and guessing, one finds that these numbers are of the form $(33\dots3)^3$. The solution is to show that

$$n_k = (\underbrace{33\dots3}_{k+1 \text{ times}})^3.$$

Notice that we can write (make sure to pay careful attention to the summation bounds, it's easy to mess up)

$$\begin{aligned} 3n_k &= \sum_{r=2k+3}^{3k+2} (10^r) + 7 \sum_{r=k+2}^{2k+1} (10^r) + 81(10^k) + \sum_{r=0}^{k-1} (10^r) \\ &= \left(1 + 7(10^{k+2}) + 10^{2k+3}\right) \frac{10^k - 1}{9} + 81(10^k) \\ &= -\frac{1}{9} + 30\frac{10^k}{9} - 300\frac{10^{2k}}{9} + 1000\frac{10^{3k}}{9} \\ n_k &= -\frac{1}{27} + 30\frac{10^k}{27} - 300\frac{10^{2k}}{27} + 1000\frac{10^{3k}}{27} \\ &= -\frac{1}{27} + 3\frac{10^{k+1}}{27} - 3\frac{10^{2k+2}}{27} + \frac{10^{3k+3}}{27} \\ &= \left(\frac{10^{k+1} - 1}{3}\right)^3 \end{aligned}$$

which is exactly the number $(\underbrace{33\dots3}_{k+1 \text{ times}})^3$ as desired. ■

Chapter 1.9 — Problem 5

Compute

$$\frac{\tan 1}{\cos 2} + \frac{\tan 2}{\cos 4} + \cdots + \frac{\tan 2^n}{\cos 2^{n+1}}.$$

Solution

We observe the following fact.

$$\begin{aligned}\tan 2^{k+1} - \tan 2^k &= \frac{\sin 2^{k+1}}{\cos 2^{k+1}} - \frac{\sin 2^k}{\cos 2^k} \\ &= \frac{\sin 2^{k+1} \cos 2^k - \sin 2^k \cos 2^{k+1}}{\cos 2^k \cos 2^{k+1}} \\ &= \frac{\sin 2^k}{\cos 2^k \cos 2^{k+1}} \\ &= \frac{\tan 2^k}{\cos 2^{k+1}}.\end{aligned}$$

Thus we can re-write the sum as

$$\sum_{k=0}^n (\tan 2^{k+1} - \tan 2^k)$$

which telescopes to $\tan 2^{n+1} - \tan 1$. ■